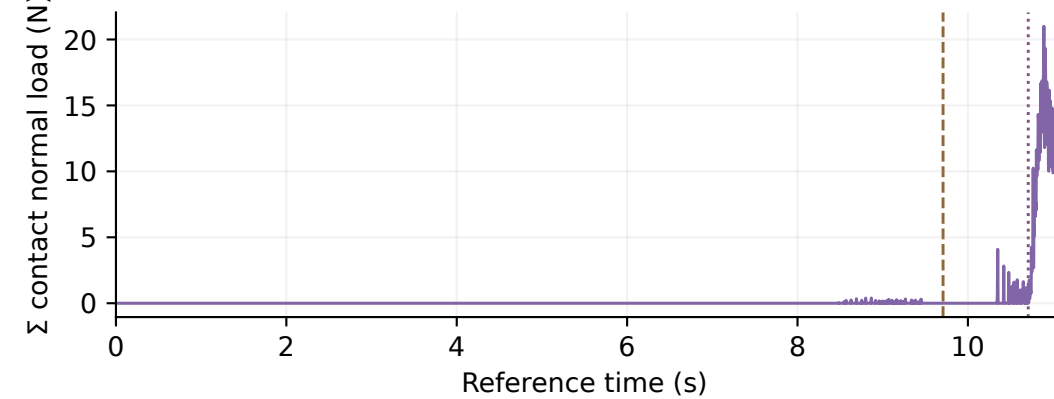
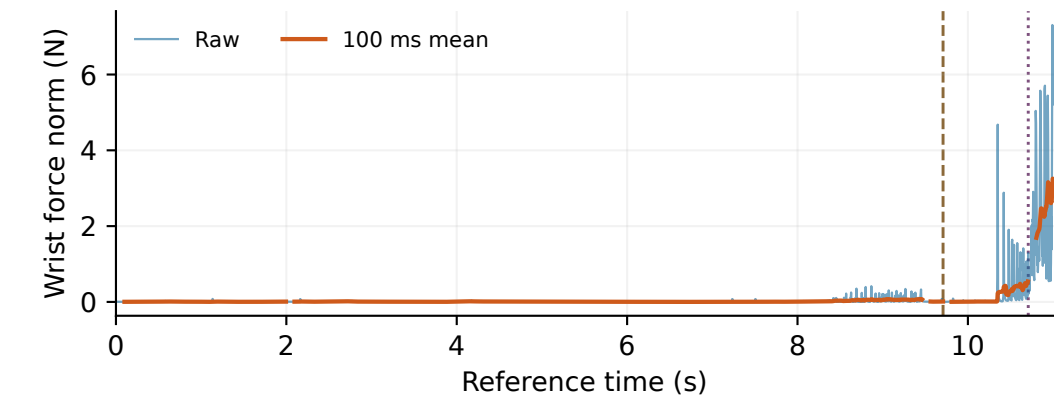
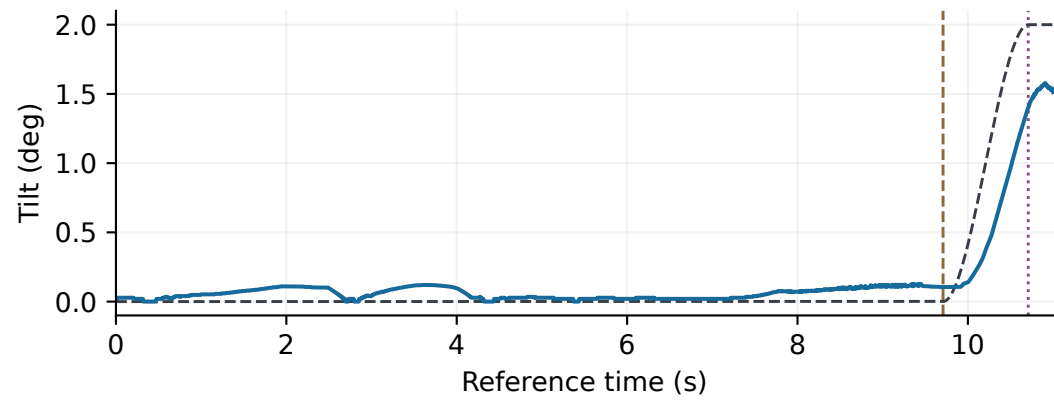
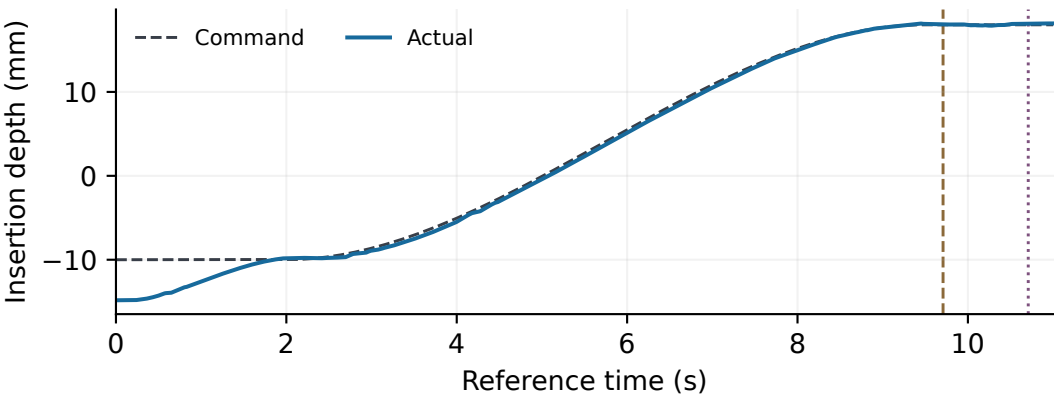


Depth 18 mm | friction (μ_s, μ_d) = (1, 1) | target tilt 2°

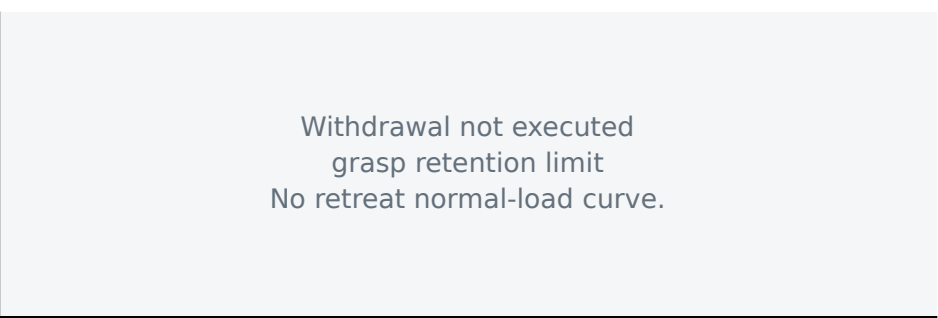
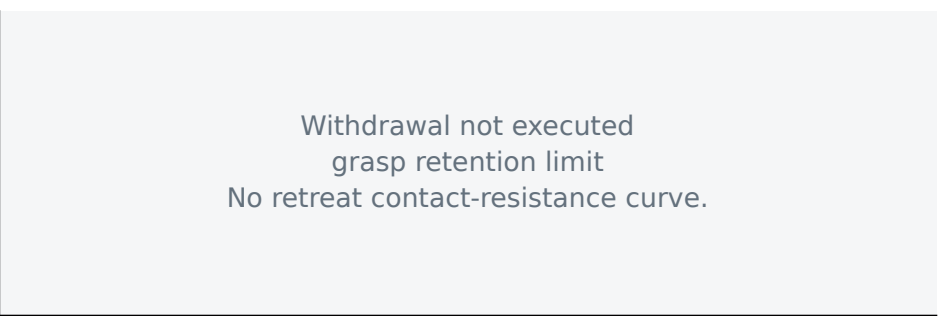
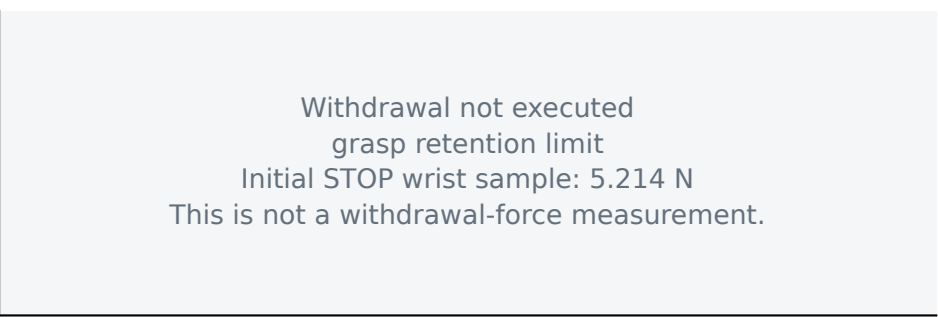
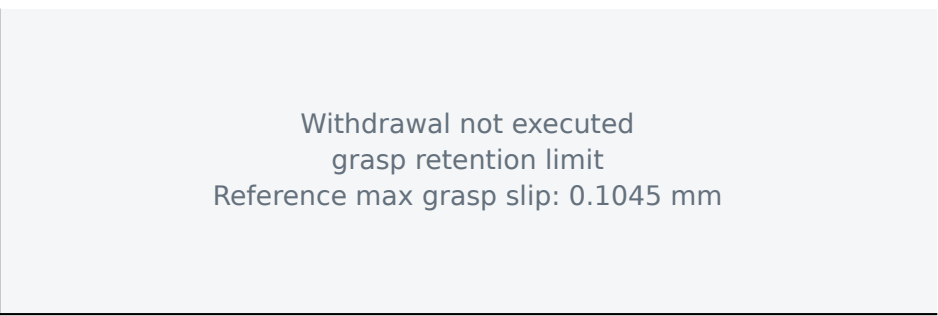
m_d18_fs100_fd100_a02 · 240 Hz · reference: grasp_lost · numerical screening: valid

Actual-depth gate / tilt start Commanded tilt ramp complete

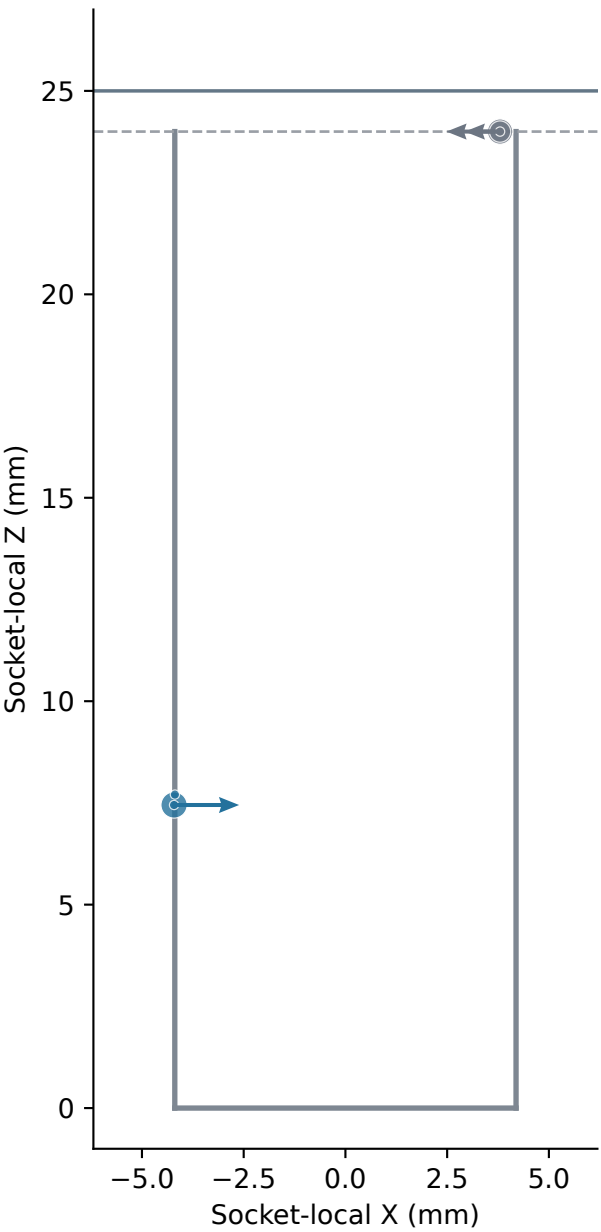
Original insertion reference



Original continuous straight recovery



Reference contact locations
max ΣN = 20.998 N at 10.892 s



7 exported loaded points
Largest point load: 8.284 N

XZ projection; Y is omitted.
Marker area and arrow length
scale with per-point normal load.
Arrows indicate force direction.

Bore guide: x = ± 4.193 mm.
Polygon facets are not drawn.
No peg mesh or deformation inferred.

Recovery result: grasp_retention_limit; recenter + align replay: reference_not_eligible (unmatched / untested). Recenter + align adjusts both XY and orientation.

Withdrawal zero is the reference endpoint depth; stop drift remains visible. Shading separates recorded recovery phases.

The 100 ms mean uses complete windows within each phase. Stop peaks are not retreat peaks. Component resistance = $\max(0, -\text{world-Z force})$.

Normal load is the sum of contact-normal magnitudes. Contact-force components and wrist reaction are different measurements; separate component peaks must not be added.

One simulated run per condition; descriptive evidence only. Missing withdrawal curves are not zero-force results.